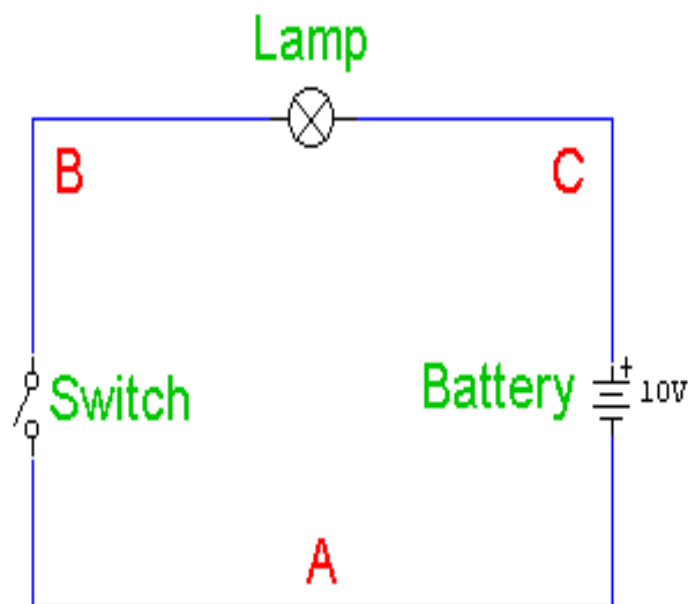


Reading Schematic diagrams

This page was created in response to all the email I receive, on how to read a schematic. Learning to read a schematic diagram, is similar to map reading. You need to know which wires connect to which component and where each wire starts and finishes. With a map book this would be equivalent to knowing your origin and destination points and which roads connect to the motorway network, etc. However schematics are a little more complicated as components need to be identified and some are polarity conscious (must be wired up in the circuit the correct way round) in order to work. You do not need to understand what the circuit does, or how it works, in order to read it, but you do need to correctly interpret the schematic. Here are some basic rules that will help with reading a diagram. Look at the circuit diagram shown below:



The blue lines represent wires and for simplicity i have labeled them as A,B,C. There are just three components here and it is easy to see where each wire starts and ends, and which components a wire is connected to. As long as the wire labelled A connects to the switch and negative terminal of the battery, wire B connects to the switch and lamp, and C connects to the lamp and the battery positive terminal then this circuit should work.

Before moving on, it is important to realise that any schematic may be drawn in a number of different ways. In Fig 1 and Fig 2 i have drawn two electrically equivalent lamp dimmer circuits, they may look very different, but in fact, if you mentally label the wires and trace them, you will see that in both diagrams each wire starts and finishes at the same components on both diagrams. The components have been labelled and so have the three terminals of the transistor.

Fig 1

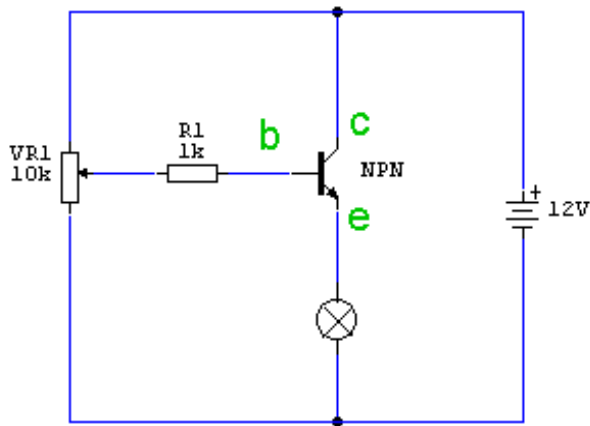
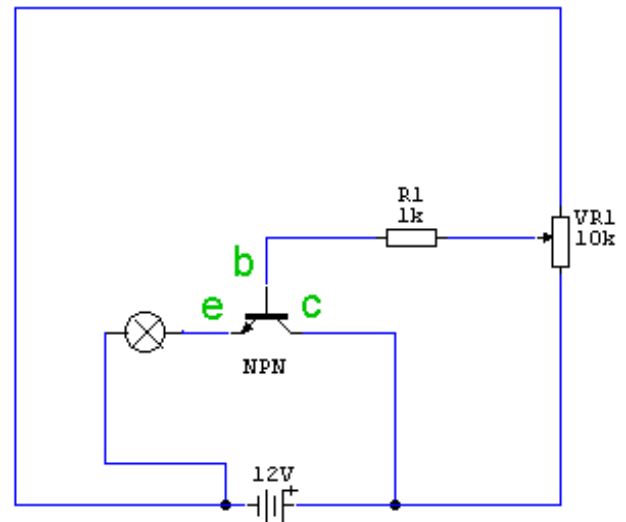


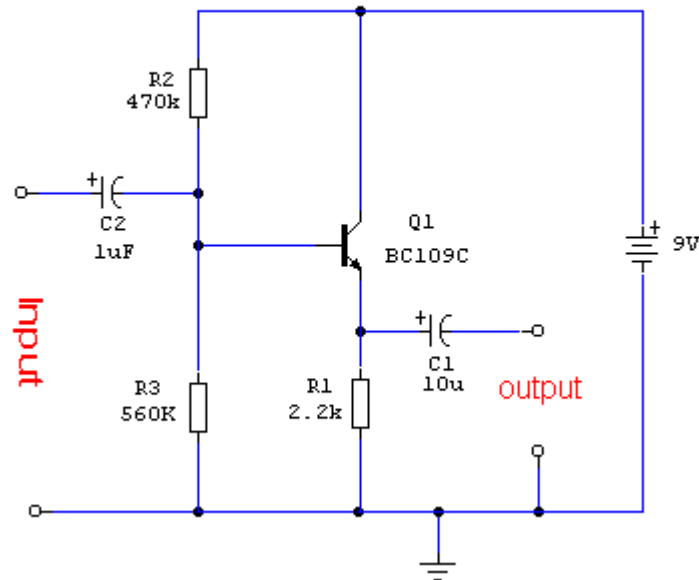
Fig 2



In Fig1 there are two wire junctions as indicated by a "dot". A wire connects from battery positive to the C (collector) terminal of the transistor, and also a wire runs from the collector terminal to one end of the potentiometer, VR1. The wires could be joined at the transistor collector, battery positive or even one end of the potentiometer, it does not matter, as long as both wires exist. Similarly, a wire runs from battery negative to the lamp, and also from lamp to the other end of VR1. The wires could be joined at the negative terminal of the battery, the lamp, or the opposite tag of VR1. In drawing Fig 1, I could have drawn the wires from the lamp and bottom terminal of VR1 back to the battery negative terminal and placed the dot there, it would still be the same. If you now look at Fig 2, you will see that one wire junction appears at the negative battery terminal, the other junction in a similar place.

Circuit Layout

Sometimes the way a circuit is wired up may compromise its performance. This is particularly important for high frequency and radio circuits, and some high gain audio circuits. See the diagram below:



Although this audio circuit has a voltage gain of less than one, wires to and from the transistor, should be kept as short as possible. This will prevent a long wire picking up radio interference or mains hum from a transformer. Also, in this circuit input and output terminals have been labelled and a common reference point or earth is indicated. The earth terminal would be connected to the chassis or metal framework of the enclosure in which this circuit is built. Many schematics contain a chassis or earth point. Generally its just to indicate the common reference terminal of the circuit, but in radio work, the earth symbol usually requires a physical connection to a cold water pipe or an earth spike buried in the soil.

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